

# Kigali Flood-Pressure Dashboard — Demo & Defense Practice

Rehearsal deck (reference only — not for submission)

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Target: a clear  $\leq 5$ -minute demo mapped to the rubric

# The 5-minute flow

Practise this order, timed

- 0:00 Framing — what flood-pressure is, and what it isn't.
- 0:30 Map + date slider — the risk map that CHANGES with rainfall.
- 1:30 Metrics + polygon overlay — agrees with the official map AND extends it.
- 2:15 Cell inspector — features → probabilities → rainfall history.
- 3:15 Per-cell transition matrix — the Markov contribution (spend time here).
- 4:00 Model & splits panel — honest metrics, no over-fitting.
- 4:45 Close — reproducible, interpretable, beats baselines.

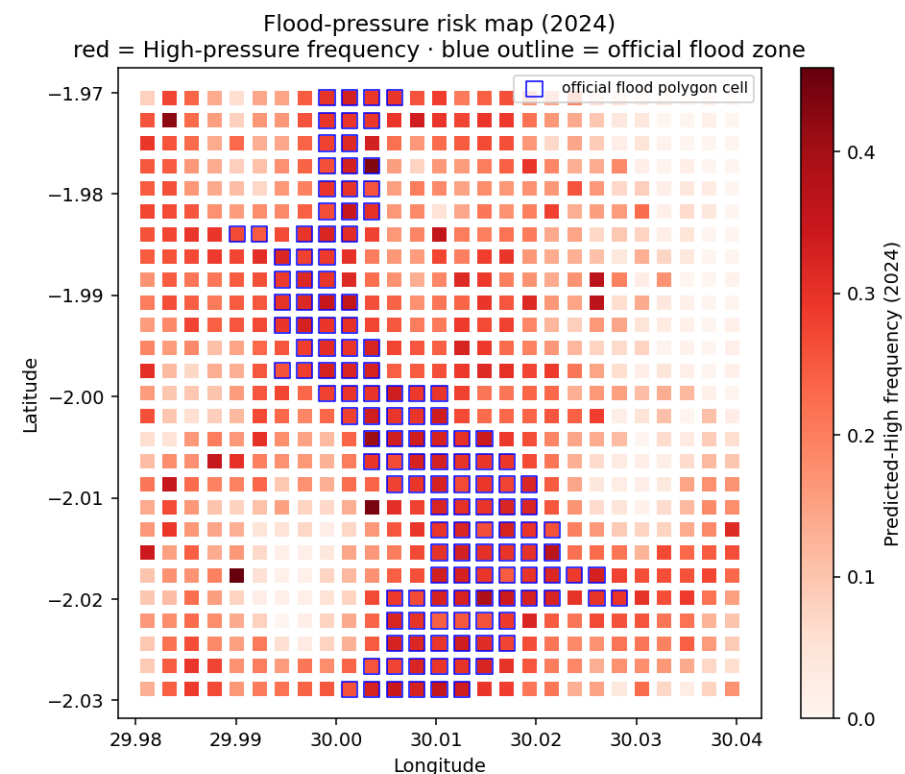
# Problem & framing (say it in 30s)

- Kigali floods after intense rain; official hazard maps are STATIC.
- They show WHERE flooding is possible — not WHEN or HOW WIDELY pressure rises.
- Our tool: daily Low / Moderate / High flood-pressure per 250 m cell.
  - Flood-pressure = derived risk state (rainfall + terrain + exposure).
  - NOT a confirmed flood event. Complements early warning, doesn't replace it.
- Real, keyless data only: ERA5 rainfall, SRTM terrain, OSM exposure, MOE polygons.

# The product — a dynamic risk map

729 cells · green Low / orange Moderate / red High

- Drag the date slider dry → wet: the red High zone grows along the river corridor.
- Hover any cell: ID, predicted class + confidence, 3-day rain, elevation.
- Three colour modes: predicted state · ground-truth label · 3-day rainfall.
- Toggle the blue official-polygon overlay to compare.



# Results — 2024 test hold-out

Targets: macro-F1  $\geq 0.75$ , High-recall  $\geq 0.80$

| Model                     | Macro-F1 | High-recall | Note                       |
|---------------------------|----------|-------------|----------------------------|
| XGBoost (deployed)        | 0.813    | 0.843       | best & shipped             |
| Random Forest             | 0.813    | 0.849       | ties; best recall          |
| Logistic Regression       | 0.750    | 0.819       | linear baseline            |
| Decision Tree             | 0.750    | 0.798       | clears the bar             |
| Rainfall-threshold (base) | 0.626    | 0.902       | high recall, low precision |
| Static-polygon (base)     | 0.324    | 0.292       | static maps inadequate     |

# Train / validation / test — no over-fitting

The exact thing your supervisor asked for

- Temporal split, NO shuffle: train 2018–2022 · validate 2023 · test 2024.
- Every cell appears in all three periods; only TIME is partitioned.
- Macro-F1 barely moves across splits:
  - XGBoost: 0.804 (train) / 0.805 (val) / 0.813 (test).
  - Random Forest: 0.813 / 0.806 / 0.813.
- Train  $\approx$  val  $\approx$  test  $\Rightarrow$  it generalises to unseen future years, not memorising.
- A random split would leak future weather into training — this is the honest test.

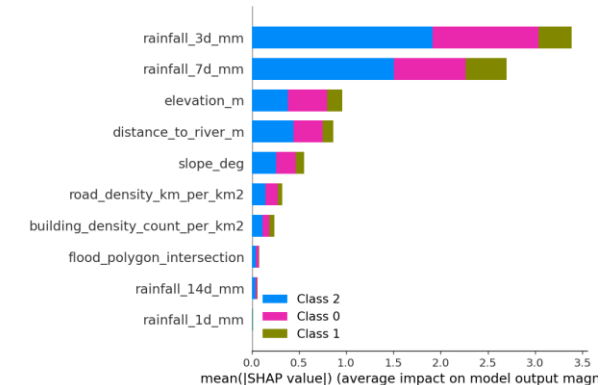
# Confusion matrix & explainability

- Diagonal = correct. Strong Low & High; error concentrates on the transitional Moderate class — off by one band, not a wild miss.
  - hydrologically sensible → auditable, not a black box.
- We optimised High-recall (0.84): a missed High day costs more than a false alarm.
- SHAP + feature importance: rainfall (3/7-day) dominates, then elevation & river.

XGBoost (2024 hold-out)

|               | Low    | Moderate | High  |
|---------------|--------|----------|-------|
| True Low      | 130325 | 11290    | 41    |
| True Moderate | 19188  | 46301    | 5491  |
| True High     | 75     | 8406     | 45697 |
|               | Low    | Moderate | High  |

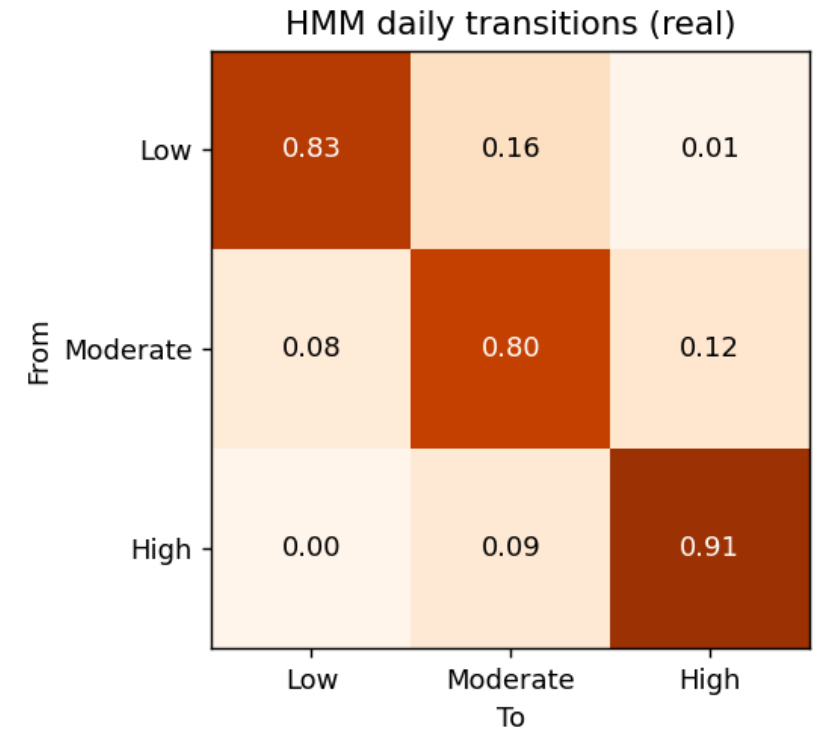
Predicted



# ★ Per-cell Markov transitions — the contribution

Not just WHAT state — what state NEXT

- For a SELECTED cell:  $P(\text{state tomorrow} \mid \text{state today})$ , from its own history.
- Strong diagonal = pressure PERSISTS after sustained rain.
- Off-diagonal Low→Moderate→High = how a cell escalates.
- '—' = a state that cell never entered (shown honestly, not faked).
- Global HMM: self-transitions  $\approx 0.83 / 0.80 / 0.91$ ; next-step acc 0.45 (chance 0.33).
  - Turns a snapshot into a short-horizon forecast a manager can act on.





# Spatial validation vs the official map — the honest result

- Official MOE zone covers only ~19% of the corridor.
- So a fixed '70% of High inside the zone' is geometrically impossible without circular labels.
- Reframed to enrichment & odds ratio:
  - Enrichment / lift: 1.60× over the 18.7% base rate.
  - Inside-vs-outside High odds: 1.85×.
- Model RE-DISCOVERS the official zone (never trained on it) AND extends the footprint.

# Rubric self-check — hit every box in the demo

| Rubric item         | Where you show it              | Number to say                      |
|---------------------|--------------------------------|------------------------------------|
| Testing Results (5) | Model panel · testing_results/ | 0.813 F1 · 10/10 tests             |
| Analysis (2)        | Map + leaderboard              | +0.19 vs rainfall, +0.49 vs static |
| Discussion          | Persistence + spatial          | 1.85× odds, agrees & extends       |
| Recommendations     | Limitations / future work      | incident validation, CHIRPS        |
| Deployment (3)      | Running app + Dockerfile       | HTTP 200 · p95 9.6 ms              |
| Demo / live app     | This walkthrough + URL         | record ≤5 min                      |

# Anticipated questions — crisp answers

- ‘Real floods?’ → No, derived pressure; incident validation is future work.
- ‘Circular label?’ → Hidden drainage + noise; F1 0.81 not 0.99.
- ‘Split / over-fit?’ → 3-way temporal; train≈val≈test (0.804/0.805/0.813).
- ‘Why XGBoost?’ → Ties RF, calibrated probs + SHAP; RF has slightly higher recall.
- ‘Why ERA5?’ → Reproducible & keyless; CHIRPS tested but fragile; Meteo Rwanda gated.
- ‘Containment only 0.30?’ → 19% coverage caps it; enrichment 1.60× / odds 1.85×.
- ‘Biggest limitation?’ → No incident register; labels are pressure, not events.

## Close (one sentence)

- A reproducible, interpretable, time-aware flood-pressure tool for Kigali —
  - beats both naive baselines on real open data,
  - agrees with and extends the official flood map,
  - and runs as a live dashboard with a passing test suite.
- Everything reruns end-to-end from one main.py. Thank you — questions welcome.